Examiner only

5. Nuclear medicine involves the use of some radioactive isotopes, that emit gamma (γ) rays, in the diagnosis and treatment of disease.

(a) Complete the following paragraph about gamma rays using the words from the box. [4]
Each word may be used once, more than once, or not at all.

particles	low	absorb	sound	strong
light	electromagnetic	penetrate	high	

Gamma rays are .... frequency ....waves. They are used in nuclear medicine because they easily .... the human body. They have .... ionising power.

(b) Information about some isotopes that emit gamma rays is given in the table below.

Isotope	Half-life
palladium-103	17.0 days
iodine-125	59.6 days
thallium-201	73.0 hours
cobalt-60	5.26 years

Use the information in the table to answer the following questions.

(i) State which isotope takes the longest time to decay. [1]

....

(ii) State what happens to the activity of thallium-201 after 73 hours. [1]

....

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(iii) In order to treat prostate cancer, palladium-103 pellets are placed directly into the prostate gland. They remain permanently in place.

I. Complete the table below. [2]

Time (days)	0	17	34	....	68
Activity of palladium-103	1	$\frac{1}{2}$	....	$\frac{1}{8}$	$\frac{1}{16}$

II. The gamma rays from the pellets are undetectable when their activity drops to $\frac{1}{32}$ of its original value. Calculate the time taken for this reduction to occur. [2]

Time = ... days

10